Project Planning Phase

Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

Date	07 March 2026
Team ID	
Project Name	**Malaria Detection using Deep Learning**

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members
Sprint-1	Dataset Preparation	USN-1	T As a developer, I want to collect microscopic blood smear images so that the model can be trained for malaria detection.	2	High	Team
Sprint-2	Data Preprocessing	USN-2	As a developer, I want to preprocess images (resize, normalize, augment) so that the dataset is suitable for deep learning models.	1	High	Team
Sprint-3	Model Developement	USN-3	As a developer, I want to build a custom CNN model to classify parasitized and uninfected cells.	2	Low	Team
Sprint-4	Model Evaluation	USN-4	As a developer, I want to evaluate the model using accuracy, precision, recall, ROC-AUC, and confusion matrix.	2	Medium	Team
Sprint-5	Prediction Module	USN-5	As a user, I want to upload a blood smear image so the system can analyze it.	1	High	Team
Sprint 6	User Interface	USN-6	As a user, I want a simple web interface to upload images and view predictions.	2	High	Team

Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{sprint\ duration}{velocity} = \frac{20}{10} = 2$$

Burndown Chart:

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

- ☐ https://www.tensorflow.org/
- ☐ https://keras.io/
- ☐ https://streamlit.io/
- ☐ **https://lhncbc.nlm.nih.gov/publication/pub9932**
- ☐ https://www.kaggle.com/datasets/iarunava/cell-images-for-detecting-malaria
- ☐ https://arxiv.org/abs/1801.04381
- ☐ https://arxiv.org/abs/1905.11946
- ☐ **https://www.who.int/teams/global-malaria-programme/reports/world-malaria-report**
- ☐ https://www.deeplearningbook.org/